

Preparation and Handling of Small-Molecule Libraries for Molecular Docking

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1 Introduction

Structure-based virtual screening (SBVS) via molecular docking has become a standard component of contemporary drug discovery workflows. In most docking campaigns, the computational protocol is only as reliable as the quality of the input the receptor structure, and equally critically, the representation and preparation of small-molecule ligands. Inadequate ligand preparation—incorrect protonation, missing tautomers, unrealistic conformations, or inconsistent valence and stereochemistry—can easily dominate the error budget of a docking calculation and lead to misleading enrichments, even if the underlying search and scoring algorithms are otherwise robust.[11, 10, 18]

The aim of this chapter is to provide a systematic and in-depth overview of methods and software for preparing and handling small molecules for docking-based virtual screening. The focus is on both theoretical underpinnings (e.g., protonation equilibria, tautomerism, conformational sampling, charge models) and practical aspects (software packages, file formats, and databases). Emphasis is placed on reproducible workflows, verification of structural correctness, and the use of curated ligand datasets and benchmarks.

After introducing the conceptual framework for ligand preparation, the chapter discusses: (i) treatment of protonation, tautomerism, and stereochemistry; (ii) conformer generation and 3D coordinate assignment; (iii) energy minimization and charge assignment; (iv) the software ecosystem for ligand preparation; (v) methods to verify structural correctness of ligands and conformations for docking; (vi) relevant small-molecule databases and benchmark datasets; and (vii) commonly used molecular structure formats with illustrative examples.

2 Conceptual Overview of Small-Molecule Preparation for Docking

2.1 Role of ligand preparation in docking

From a physical perspective, docking attempts to approximate the binding free energy

$$\Delta G_{\text{bind}} \approx -RT \ln K_d,$$

where K_d is the dissociation constant, R is the gas constant, and T is the absolute temperature. A docking score is a rough surrogate for ΔG_{bind} , and is computed on a particular protein–ligand complex geometry.[18] For this score to be meaningful, the docked pose must correspond to a chemically realistic ligand in an accessible protonation, tautomeric, and conformational state.

Ligand preparation therefore aims to:

1. **Standardize connectivity and valence:** ensure correct bond orders, aromaticity perception, and formal charges are consistent with chemical intuition and experimental $\text{p}K_a$ /redox state.
2. **Generate relevant protonation and tautomeric states:** enumerate or select those states likely populated at the target pH and environment.
3. **Resolve stereochemistry and isomerism:** define chiral centers and E/Z double-bond stereochemistry, enumerating alternatives if necessary.
4. **Generate low-energy 3D conformations:** sample accessible conformational space so that docking can identify plausible binding modes.
5. **Assign force-field parameters and partial charges:** provide the parameters required by the docking engine’s scoring function.
6. **Filter and quality-control:** remove impossible or highly strained structures, salts, and problematic chemotypes.

Extensive benchmarking has shown that systematic attention to these steps can markedly improve virtual screening enrichment factors, sometimes by factors of two or more.[11, 1]

2.2 Typical workflow

A typical ligand preparation workflow for docking-based virtual screening can be outlined as:

1. Input 1D/2D structures (SMILES, SDF, etc.) and perform basic sanitization (valence checks, standardization).
2. Strip salts and solvents; select the main organic component.
3. Predict and enumerate relevant protonation states and tautomers at specified pH (e.g., 7.4 ± 1).
4. Enumerate undefined stereocenters and E/Z isomers, within user-defined limits.
5. Generate 3D coordinates for each state and perform conformational sampling.
6. Perform restrained energy minimization using a small-molecule force field.
7. Assign partial charges and any special atom types needed by the docking program.
8. Export to the docking engine’s preferred file format (e.g., MOL2, PDBQT, Maestro).
9. Verify geometry and chemistry with automated and manual checks.

Software such as Open Babel,[15] RDKit (with ETKDG-based embedding),[16] OMEGA,[4] and Gypsum-DL[17] implement many of these steps in open-source form, while commercial suites (Schrödinger, OpenEye, MOE, Discovery Studio) provide additional proprietary functionality.

3 Representation and Basic Processing of Small Molecules

3.1 1D, 2D, and 3D representations

Small molecules may be initially represented as:

- **1D line notations:** e.g., SMILES,[20] InChI.[5] These encode molecular graphs as strings, convenient for storage and substructure/similarity search.
- **2D connection tables:** e.g., MDL MOL/SDF, which store atom types, bond orders, and sometimes 2D coordinates.
- **3D coordinate files:** e.g., MOL2, PDB, Maestro, which include 3D coordinates suitable for docking and molecular mechanics.

Docking requires 3D geometries, but upstream ligand sources (ChEMBL, ZINC, PubChem) are often provided as 1D SMILES or 2D SDF formats, necessitating generation of 3D coordinates and conformers.[8, 3, 9]

3.2 Sanitization: valence, aromaticity, and charge checking

Most cheminformatics toolkits implement a “sanitization” step that verifies chemical plausibility:

- **Valence checks:** ensure that valence counts of atoms conform to allowed patterns (e.g., tetravalent carbon, trivalent nitrogen in amines).
- **Aromaticity perception:** apply a model (e.g., Hückel-like rules or pre-tabulated aromatic atom types) to label aromatic rings.
- **Bond order perception:** infer missing bond orders from connectivity and valence rules.
- **Charge balancing:** ensure total charge is consistent with valence and, where possible, with expected pK_a .

In RDKit, for example, sanitization is performed via `Chem.SanitizeMol`, which checks valence, aromaticity, conjugation, and ring information, raising errors or warnings when invalid structures are encountered. Open Babel provides analogous functionality, including detection of impossible valence and inconsistent aromaticity.[15]

Sanitization should be done *before* any optimization or docking-related processing, because errors at this stage can invalidate the entire pipeline.

3.3 Standardization and normalization

Standardization aims to convert diverse input representations into a consistent internal representation. Common steps include:

- **Salt stripping:** remove counterions (e.g., chloride, sodium) and solvents (e.g., water, DMSO) using fragment-based rules.
- **Normalization of functional groups:** convert alternative representations (e.g., nitro as $\text{N}^+(\text{=O})[\text{O}^-]$) into a canonical pattern.
- **Neutralization or pH normalization:** adjust charges on strong acids/bases to a standard reference pH or to a neutral form.
- **Stereo cleanup:** remove or set undefined chiral centers when the input stereo is ambiguous.

Standardization libraries (e.g., RDKit’s `rdMolStandardize`, commercial standardization tools) encapsulate rules curated from medicinal chemistry practice.

4 Protonation States, Tautomerism, and Stereochemistry

4.1 Protonation equilibria and the Henderson–Hasselbalch equation

The protonation state of ionizable groups (e.g., carboxylic acids, amines) is governed by acid–base equilibria. For a monoprotic acid HA,



the ratio of deprotonated to protonated species at equilibrium is given by the Henderson–Hasselbalch equation:

$$\text{pH} = \text{p}K_a + \log_{10} \frac{[\text{A}^-]}{[\text{HA}]}.$$

Rearranging,

$$\frac{[\text{A}^-]}{[\text{HA}]} = 10^{\text{pH} - \text{p}K_a}.$$

For a given pH and known $\text{p}K_a$, this allows the computation of the fractional populations of each protonation microstate. For multiple ionizable

centers, microstate populations can be computed from a micro- K_a or micro- pK_a model; in practice, software such as Epik, ChemAxon pK_a predictors, or Dimorphite-DL (used within Gypsum-DL) implement empirical or semi-empirical models to enumerate and score protonation microstates.[17, 11]

In docking, it is rarely sufficient to consider a single protonation state; instead, all states within a reasonable free energy window (often a few kcal/mol at the working pH) should be generated. The Boltzmann weight of each state i is

$$p_i = \frac{e^{-\beta G_i}}{\sum_j e^{-\beta G_j}}, \quad \beta = \frac{1}{k_B T},$$

where G_i is the free energy of microstate i . At typical screening pH, states whose populations are below 1–5 % may be neglected, but borderline cases (e.g., histidines, anilines) can strongly influence binding modes.

4.2 Tautomerism

Many heterocycles and functional groups exhibit tautomerism (e.g., keto–enol, amide–imidic acid, lactam–lactim). Tautomers often differ in hydrogen placement and formal charges, with potentially large effects on hydrogen bonding and electrostatics in the binding site. Automatic tautomer enumeration proceeds by defining substructure-based transformation rules (SMARTS patterns) that map between tautomeric forms.

In practice:

- Enumerators generate all possible tautomers under a given set of rules.
- Scoring functions or empirical rules estimate relative stability (e.g., favoring amide over imidic acid forms).
- A subset of low-energy tautomers is selected for docking, sometimes weighted by estimated populations.

Tools such as Gypsum-DL explicitly enumerate tautomers (and other isomeric forms) and produce 3D structures for each selected form.[17] [cite:135] Commercial programs (e.g., LigPrep/Epik, MOE Wash) implement proprietary tautomer generators and scoring schemes.[11]

4.3 Stereochemistry and E/Z isomerism

Stereochemical ambiguities are common in large virtual libraries:

- Chiral centers may be unspecified (e.g., input from generic SMILES).

- E/Z configuration of alkenes may be unknown or partially constrained.
- Atropisomerism and conformational chirality are usually not explicitly encoded.

A general strategy is to:

1. Keep experimentally known stereochemistry fixed (from curated databases like ChEMBL or PDB ligands).
2. Enumerate all reasonable stereoisomers for undefined centers, but cap the total number of isomers per molecule to control combinatorial explosion.
3. For flexible or unassigned stereocenters (e.g., tertiary alcohols in fragment libraries), consider whether stereochemistry is relevant to the docking hypothesis.

Gypsum-DL, for example, can enumerate R/S stereocenters, E/Z isomers, and ring conformers, within user-defined limits, for each input molecule.[17]

5 3D Coordinate Generation and Conformational Sampling

5.1 Systematic vs. stochastic conformer generation

The goal of conformer generation is to produce an ensemble of low-energy 3D geometries that approximate the accessible conformational space of a molecule in solution, while respecting chirality and other stereochemical constraints. Broadly, two classes of methods are used:[4, 16]

- **Systematic, rule-based sampling:** Rotatable bonds are identified, and torsions are sampled from libraries of preferred angles; combinatorial explosion is controlled by pruning and energy filters.
- **Stochastic sampling:** Monte Carlo or distance-geometry-based methods sample conformational space randomly or quasi-randomly, often combined with local minimization.

[cite:18][cite:54]

5.2 OMEGA: systematic rule-based conformer generation

OMEGA (OpenEye) is a widely used systematic conformer generator for small molecules, and has become a de facto standard in many docking workflows.[4] Its algorithm operates in three phases:

1. **Fragment-based initial structure assembly:** Molecules are decomposed into fragments at rotatable bonds; fragment libraries derived from crystallographic data provide 3D templates.
2. **Exhaustive torsion enumeration:** Rotatable bonds are sampled at discrete torsion angles from a knowledge-based torsion library. This generates a potentially large raw conformer set.
3. **Pruning by geometry and energy:** Conformers that are too similar (RMSD threshold) or too high in strain energy are discarded to yield a diverse low-energy ensemble.

A simple measure of structural similarity between two conformers is the heavy-atom root-mean-square deviation (RMSD),

$$\text{RMSD} = \sqrt{\frac{1}{N} \sum_{i=1}^N \left\| \mathbf{r}_i^{(\text{ref})} - \mathbf{r}_i^{(\text{fit})} \right\|^2},$$

where N is the number of matched atoms, and $\mathbf{r}_i^{(\text{ref})}$ and $\mathbf{r}_i^{(\text{fit})}$ are their coordinates after optimal superposition.

OMEGA has been extensively validated against high-quality crystal structures from the PDB and Cambridge Structural Database, demonstrating high success rates in reproducing experimental ligand conformations within low RMSD thresholds.[4]

5.3 Distance geometry and ETKDG (RDKit)

Distance geometry (DG) methods construct 3D coordinates by embedding a matrix of pairwise distance bounds between atoms.[16] Briefly:

1. Define lower and upper bounds $d_{ij}^{\min}$, $d_{ij}^{\max}$ for each atom pair i, j , based on covalent geometry and topological distance.
2. Apply triangle inequality constraints to make the bounds matrix metrically consistent.

3. Sample a distance matrix from within the bounds and perform metric matrix embedding (eigen-decomposition) to obtain coordinates.
4. Optionally refine the coordinates with a force field.

The ETKDG (Experimental-Torsion Knowledge Distance Geometry) method extends basic DG by incorporating empirical torsion angle preferences derived from crystallographic databases, and by applying additional constraints for ring systems.[16] Implemented in RDKit, ETKDG produces conformers that both reproduce experimental torsion distributions and reduce the need for extensive post-embedding minimization. [cite:54]

For macrocycles and very flexible molecules, ETKDG variants and specialized sampling strategies are needed to adequately explore conformational space.[16]

5.4 Energy minimization and strain filtering

Conformers produced by OMEGA, ETKDG, or other methods are often subjected to a brief energy minimization using a molecular mechanics force field such as MMFF94, UFF, or GAFF. The typical nonbonded energy expression consists of Lennard–Jones and Coulomb terms:

$$E = \sum_{i < j} \left[4\varepsilon_{ij} \left(\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right) + \frac{q_i q_j}{4\pi\varepsilon_0\varepsilon_r r_{ij}} \right],$$

where r_{ij} is the interatomic distance, ε_{ij} and σ_{ij} are Lennard–Jones parameters, and q_i are partial charges.

Strain energy thresholds are applied to filter out highly strained conformers that would be unlikely in solution and could bias docking scores. In practice, these thresholds are empirical (e.g., 10–15 kcal/mol above the global minimum for a small molecule), but they should be chosen carefully: overzealous filtering might remove conformers relevant for binding, especially in tight or distorted pockets.

6 Partial Charge Assignment and Force-Field Parameters

6.1 Charge models for docking

Docking scoring functions typically include an electrostatic term that depends on partial atomic charges. Common small-molecule charge models include:

- **Empirical charge equilibration:** e.g., Gasteiger–Marsili charges, fast and widely available in Open Babel, RDKit, and many docking codes.
- **Semiempirical-based schemes:** e.g., AM1-BCC charges, which fit bond-charge corrections to reproduce HF/6-31G* electrostatic potentials.[11]
- **RESP/ESP-fitted charges:** derived from higher-level QM calculations; common in MD force fields (e.g., GAFF/AMBER), but expensive for large libraries.

The choice of charge model must be compatible with the docking engine and, ideally, with the force field used for both ligand and receptor. Some protocols (e.g., Glide/SP) use internally parameterized charges and do not expose them to the user, while others (e.g., AutoDock4, Vina) require explicit charge assignment in the input (Gasteiger or similar).

6.2 Parameter assignment tools

For general organic molecules, GAFF (AMBER) and CGenFF (CHARMM) are commonly used to assign force-field parameters for downstream MD, but for docking, simpler parameterizations are often sufficient. Open-source tools such as Open Babel and RDKit can assign basic atom types and MMFF94 parameters; commercial suites provide more extensive coverage of medicinal chemistry space.

7 Software Ecosystem for Ligand Preparation

7.1 Open-source tools

Open Babel. Open Babel is a widely used open-source cheminformatics toolkit that supports interconversion between over 100 file formats, basic geometry optimization, charge assignment, and 3D coordinate generation.[15] It is frequently used as a glue layer in ligand preparation pipelines, especially for converting between SMILES, SDF, MOL2, and PDBQT formats.

RDKit. RDKit is a modern open-source cheminformatics library with robust functionality for molecule sanitization, standardization, ETKDG-based conformer generation, substructure search, and basic property calculations.[16] Its Python API makes it an attractive choice for custom ligand preparation scripts.

Gypsum-DL. Gypsum-DL is an open-source program specifically designed to prepare small-molecule libraries for structure-based virtual screening.[17] Given SMILES or SDF input, Gypsum-DL:

- Enumerates ionization states (via Dimorphite-DL) over a user-defined pH range.
- Enumerates tautomers, R/S stereocenters, E/Z isomers, and ring conformers.
- Generates 3D coordinates using RDKit and filters redundant conformers.

Gypsum-DL has been shown to improve docking pose prediction in combination with PDBbind-derived complexes.[17]

Other tools. Additional open-source utilities include:

- **Cinfony:** a Python wrapper that unifies interfaces to Open Babel, RDKit, and CDK.[?]
- **Dimorphite-DL:** pH-dependent protonation state predictor, often used with Gypsum-DL.
- **OpenEye free academic tools:** limited but valuable for conformer and charge generation.

7.2 Commercial ligand preparation suites

Schrödinger LigPrep/Epik. LigPrep, coupled with Epik for pK_a prediction, is widely used in industry for systematic ligand preparation.[11] It performs:

- State enumeration (protonation and tautomeric) across pH.
- Stereoisomer enumeration.
- OPLS-based minimization and partial charge assignment.
- Export of docking-ready Maestro or MOL2 files.

The impact of various settings in these tools on virtual screening enrichments has been systematically explored.[11]

OpenEye QUACPAC/OMEGA. In addition to OMEGA for conformer generation, OpenEye provides QUACPAC for charge state and partial charge assignment, and FILTER for removal of undesirable chemotypes.[4] These tools are heavily used in industrial pipelines.

MOE, Discovery Studio, others. Both MOE (Chemical Computing Group) and Discovery Studio (BIOVIA) implement comprehensive ligand preparation workflows (salt stripping, neutralization, protonation, stereochemistry, conformer generation) integrated with their docking engines.

8 Verification and Quality Control of Ligand Structures

8.1 Chemistry and valence validation

Key checks include:

- **Valence violations:** atoms with impossible valence (e.g., pentavalent carbon, pentavalent neutral nitrogen) or hypervalent halogens without appropriate hypervalent models.
- **Aromaticity and ring perception:** ensure aromatic rings are planar and correctly labeled; avoid inconsistent aromatic/antiaromatic assignments.
- **Formal charge sanity:** check for unexpected charge separation, zwitterions, or net charges inconsistent with known chemistry.

Automated sanitization in RDKit or Open Babel catches many such problems, but additional curated filters (e.g., PAINS, reactive groups) are commonly applied in drug-like libraries.

8.2 Geometric quality: bond lengths, angles, and torsions

8.2.1 Bond lengths and angles

Bond lengths and angles should be within reasonable ranges for the corresponding bond orders and atom types. Force-field minimization tends to regularize these, but conformations generated purely by distance geometry may exhibit distorted geometries if poorly refined.[16] Visual inspection of a

subset of ligands and targeted checks for pathologies (e.g., extremely short nonbonded distances, highly bent triple bonds) are recommended.

8.2.2 Ring conformations and planarity

For aromatic and conjugated rings (e.g., benzene, heteroaromatics, amides), planarity is a useful quality criterion. Deviations from planarity can be quantified by measuring the root-mean-square deviation of ring atoms from the best-fit plane:

$$\text{RMSD}_{\text{plane}} = \sqrt{\frac{1}{N} \sum_{i=1}^N d_i^2},$$

where d_i is the perpendicular distance of atom i from the best-fit plane. Large deviations (e.g., $> 0.1\text{--}0.2\text{ \AA}$ for benzene) suggest geometric artifacts.

Macrocycles and flexible rings may legitimately adopt nonplanar conformations, but small aromatic rings and amide linkages should be close to planar in starting conformers.

8.3 Internal strain and unrealistic intramolecular interactions

Excessive internal strain—e.g., extremely close nonbonded contacts, twisted amides, eclipsed torsions—may artificially stabilize or destabilize certain docking poses when combined with the receptor. Minimization with a physically sensible force field mitigates many such issues, but energy-based filters can be more systematic.

If an approximate relative strain energy E_i of conformer i can be computed (e.g., MMFF94), a Boltzmann-like weight can be used to filter:

$$w_i = \frac{e^{-\beta E_i}}{\sum_j e^{-\beta E_j}}.$$

Conformers whose weights fall below a threshold may be discarded from the docking ensemble.

8.4 Matching prepared ligands to experimental structures

For retrospective validation and method development, it is valuable to compare prepared ligand conformations and protonation states with crystallographic reference structures from PDB, PDBbind, or curated sets (DUD-E,

DEKOIS 2.0). RMSD between prepared and crystallographic conformers, as well as agreement in hydrogen-bond donor/acceptor patterns and formal charges, provide sanity checks.[4? , 19]

Irwin and co-workers have emphasized the importance of verifying that docking scores and poses correspond to chemically reasonable structures, arguing that simple metrics like RMSD to known structures and inspection of key interactions are essential to ensure docking is “right for the right reasons.”[?]

8.5 Pose-level verification for docking

Although this chapter focuses on pre-docking preparation, pose-level checks during or after docking are closely related. Useful diagnostics include:

- **Ligand strain on binding:** compare energy of docked conformer in isolation vs. its lowest-energy solution conformer.
- **Clashes:** detect steric clashes between ligand and protein atoms beyond what the scoring function penalizes.
- **Interaction patterns:** ensure hydrogen bonds, salt bridges, and hydrophobic contacts are physically plausible, given prepared protonation states.

Benchmarks such as DUD-E and DEKOIS 2.0 provide frameworks to evaluate how ligand preparation and pose filtering impact early enrichment and enrichment factors.[14?]

9 Databases and Datasets for Docking-Based Virtual Screening

9.1 ZINC: commercially available compounds for virtual screening

ZINC (“ZINC Is Not Commercial”) is a free database of commercially available small molecules prepared specifically for virtual screening.[8] It provides:

- Ready-to-dock 3D structures, often in multiple protonation and tautomeric states.
- Multiple file formats (SMILES, MOL2, SDF, DOCK flexibases).

- Annotated subsets (e.g., lead-like, fragment-like, drug-like).

[cite:20][cite:23]

The database has grown from hundreds of thousands of compounds in early versions to tens of millions of purchasable molecules in recent releases, and remains a primary source for structure-based virtual screening libraries.

9.2 ChEMBL: bioactivity-centric compound database

ChEMBL is an open, manually curated bioactivity database that aggregates binding, functional, and ADMET data for over a million small molecules and thousands of protein targets.[3] For docking, ChEMBL is primarily:

- A source of known actives and inactives for specific targets, useful for training, validation, and benchmarking.
- A source for constructing focused libraries around chemotypes of interest.

ChEMBL is distributed with compound structures (typically as standardized SMILES or SDF), which can be fed into ligand preparation pipelines. Efforts to curate stereochemistry, salt forms, and activity data quality make it attractive for structure-based modeling.

9.3 PubChem: large public chemical information repository

PubChem, maintained by NCBI, is a large public chemical database with substance, compound, and bioassay collections, encompassing hundreds of millions of substance records and tens of millions of unique structures.[9] It offers:

- Open access to chemical structures and associated bioactivity data.
- Powerful search tools (substructure, similarity, property filters).
- Links to genes, proteins, and pathways.

For docking, PubChem is often used to:

- Retrieve known ligands and analogs of a target.
- Build custom libraries around scaffolds or pharmacophores of interest.

Because PubChem structures originate from many depositors, rigorous ligand preparation and standardization are essential before docking.

9.4 PDB and PDBbind: protein–ligand complexes and binding affinities

The Protein Data Bank (PDB) is the primary repository of macromolecular structures, including many protein–ligand complexes.[2] These complexes define binding site geometries, water networks, and ligand poses used for docking validation and training.

The PDBbind database builds on PDB by collecting experimentally measured binding affinities (K_d , K_i , IC_{50}) for protein–ligand complexes and organizing them into curated sets.[19] PDBbind’s “refined set” is widely used to:

- Benchmark docking and scoring functions.
- Train machine-learning models of binding affinity.
- Validate ligand preparation and conformer generation against crystallographic ligands.

9.5 DUD-E: directory of useful decoys, enhanced

DUD-E is a benchmark library designed to evaluate docking and virtual screening methods.[14] It comprises:

- 102 protein targets with known active ligands.
- Property-matched decoys: compounds that resemble actives in simple physicochemical properties but are topologically dissimilar and presumed inactive.

DUD-E allows assessment of enrichment metrics (e.g., ROC-AUC, early enrichment at 1%) and sensitivity to ligand preparation. Mysinger *et al.* showed that improved decoy selection and more diverse ligands in DUD-E enhance the discriminatory power of docking benchmarks.[14]

9.6 DEKOIS 2.0: challenging docking benchmark sets

DEKOIS 2.0 provides challenging docking benchmark sets built from BindingDB bioactivity data for 81 diverse targets.[1] It emphasizes:

- Rigorous physicochemical matching of decoys to actives, including charge.
- Elimination of latent actives among decoys.

[cite:84][cite:81]

Bauer *et al.* used DEKOIS 2.0 to investigate the impact of ligand and protein preparation (e.g., protonation states, conformer generation) on virtual screening performance, highlighting how different preparation protocols can significantly alter enrichment.[1, 7]

9.7 Other datasets and ML-focused docking resources

Beyond classical benchmarks, more recent datasets geared towards ML-based docking and scoring, such as Smiles2Dock, provide large-scale collections of docked poses and scores for millions of ligand–protein pairs.[13] These resources encourage development of docking surrogates and hybrid ML–physics approaches, but require careful consideration of the underlying ligand preparation methods used in dataset generation.

10 Molecular Structure File Formats

10.1 SMILES: simplified molecular-input line-entry system

SMILES, introduced by Weininger,[20, 22, 21] is a line notation for encoding molecular graphs as ASCII strings. It supports branching, ring closures, stereochemistry, and charges. Examples:

- Ethanol: CCO or OCC
- Benzene: c1ccccc1
- Protonated amine: C[NH3+]

[cite:106]

SMILES is compact and human-readable, and is ubiquitous in cheminformatics databases and ligand preparation workflows. However, it does not contain 3D coordinates; conversion to 3D requires embedding via tools like RDKit or Open Babel.

10.2 InChI: IUPAC International Chemical Identifier

InChI is a nonproprietary identifier developed under IUPAC to provide a unique, canonical label for a defined chemical structure.[5, 12, 6] InChI strings encode layers (connectivity, hydrogen, charge, stereochemistry, isotopes), and

InChIKey is a hashed, fixed-length representation suitable for indexing and web search.

InChI is ideal for database linking and deduplication, but is less convenient for substructure or similarity search than SMILES. It is not directly used as a docking input format but is often used to harmonize ligand sets across databases (e.g., PDBbind–ChEMBL cross-mapping).

10.3 SDF/MOL: MDL connection tables

MDL MOL files (single molecules) and SDF (Structure Data File, multiple molecules) are widely used connection table formats. An SDF record for ethanol might look like:

```
Ethanol
  RDKit          2D

  3  2  0  0  0  0          999 V2000
    0.0000    0.0000    0.0000 C   0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
    1.5400    0.0000    0.0000 C   0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
    2.0450    1.2300    0.0000 O   0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
  1  2  1  0
  2  3  1  0
M  END
>  <ID>
ETHANOL

>  <SMILES>
CCO
```

SDF allows arbitrary associated data fields (e.g., ID, SMILES, properties) and is a common interchange format for docking libraries.

10.4 MOL2: Tripos MOL2 format

MOL2 is a richer 3D format that stores atom types, charges, and substructure information. A simplified MOL2 for ethanol might include:

```
@<TRIPOS>MOLECULE
ETHANOL
3 2 1 0 0
```

```
SMALL
NO_CHARGES
```

```
@<TRIPOS>ATOM
1 C1  0.0000  0.0000  0.0000 C.3 1 ETH 0.000
2 C2  1.5400  0.0000  0.0000 C.3 1 ETH 0.000
3 O3  2.0450  1.2300  0.0000 O.3 1 ETH 0.000
```

```
@<TRIPOS>BOND
1 1 2 1
2 2 3 1
```

MOL2 is commonly used by docking programs such as DOCK, GOLD, and FRED, and is well supported by Open Babel and RDKit.

10.5 PDB and PDBQT

PDB. The PDB format was originally developed for macromolecular structures.[2] Small-molecule ligands in PDB files are represented using HETATM records. A minimal ligand block:

HETATM	1	C1	LIG A	1	10.123	12.345	5.678	1.00	20.00	C
HETATM	2	C2	LIG A	1	11.500	12.000	5.500	1.00	20.00	C
HETATM	3	O3	LIG A	1	12.000	13.200	5.800	1.00	20.00	O

For docking, PDB is often converted to other formats (e.g., MOL2, PDBQT) that store explicit bond orders and charges more reliably than legacy PDB conventions.

PDBQT. PDBQT is an extension of PDB used by AutoDock and AutoDock Vina, adding partial charges (Q) and atom types (T) for the AutoDock force field. Bonds and rotatable torsions are encoded in separate sections. For example:

```
REMARK  Autodock PDBQT file
ROOT
ATOM      1  C1  LIG      1      10.123  12.345   5.678  0.00  C
ATOM      2  C2  LIG      1      11.500  12.000   5.500  0.00  C
BRANCH 1 2
ATOM      3  O3  LIG      1      12.000  13.200   5.800 -0.50  OA
ENDBRANCH 1 2
ENDROOT
TORSDOF 1
```

Conversion from SDF/MOL2 to PDBQT is typically done via AutoDockTools scripts or wrappers in Open Babel (`obabel -i sdf -o pdbqt`).

10.6 Maestro (MAE) and other proprietary formats

Schrödinger’s Maestro (MAE) format and other proprietary binary/ASCII formats (e.g., SD Encrypted) store rich structural and property information. Although not human-readable, they are well supported within their respective ecosystems and often serve as the internal canonical representation during ligand preparation.

10.7 File format considerations for docking pipelines

Choice of file format for docking involves trade-offs:

- **SMILES**: compact, ideal for initial library storage and substructure search; requires 3D embedding before docking.
- **SDF**: flexible and widely supported; stores multiple molecules and arbitrary properties.
- **MOL2**: stores atom types and charges, often preferred for docking.
- **PDBQT**: required by AutoDock/Vina; less general but essential in those pipelines.

Open Babel, RDKit, and other toolkits are typically used to interconvert between these formats.[15]

11 Putting It All Together: A Practical Ligand Preparation Workflow

This section outlines a concrete, fully open-source workflow for preparing a library of SMILES strings for docking with a generic docking engine that accepts MOL2 or PDBQT inputs.

11.1 Step 1: Input and sanitization

1. Start from a curated SMILES library, e.g., a subset of ZINC or ChEMBL filtered for basic drug-like properties.

2. Use RDKit to parse each SMILES into a molecule object and perform sanitization:
 - Catch valence errors and discard problematic entries.
 - Apply standardization (functional-group normalization, salt stripping) using predefined rules.

11.2 Step 2: Protonation and tautomer enumeration

1. For each sanitized molecule, use Gypsum-DL to enumerate protonation states and tautomers across $\text{pH } 7.4 \pm 1$:
 - Limit the number of protonation/tautomeric states per molecule (e.g., up to 16).
 - Reject highly unstable or chemically unreasonable states.
2. Alternatively, use RDKit plus an external $\text{p}K_a$ estimator or ChemAxon tools if available.

11.3 Step 3: Stereochemistry and isomer enumeration

1. Within Gypsum-DL or a custom RDKit script, enumerate undefined stereocenters and E/Z isomers subject to:
 - Restrictions on the maximum number of stereoisomers (e.g., up to 32).
 - Rules to avoid redundant enumeration of centers where chirality is irrelevant (e.g., pseudoasymmetric centers).

11.4 Step 4: 3D conformer generation

1. For each enumerated structure, run RDKit’s ETKDG to generate an initial ensemble of conformers (e.g., up to 20 per structure).
2. Optionally, supplement or validate RDKit-generated conformers with OMEGA for a subset of molecules, particularly macrocycles or highly flexible chemotypes.[4, 16]

11.5 Step 5: Energy minimization and strain filtering

1. Minimize each conformer with MMFF94 via RDKit or Open Babel.
2. Compute relative strain energies and discard conformers exceeding a threshold (e.g., 10 kcal/mol above the minimum).
3. Optionally cluster conformers based on heavy-atom RMSD to ensure diversity.

11.6 Step 6: Charge assignment

1. If targeting AutoDock/Vina, assign Gasteiger partial charges using Open Babel or AutoDockTools.
2. For other docking engines, use MMFF94 or AM1-BCC charges if supported, or the engine's internal charge model.

11.7 Step 7: Export to docking format

1. Export per-structure conformers to SDF or MOL2 using Open Babel (`obabel`).
2. For AutoDock/Vina, convert MOL2/SDF to PDBQT and define rotatable bonds.

11.8 Step 8: Quality control

1. Randomly sample a few hundred ligands and visually inspect:
 - Protonation states and tautomers (do they match expected chemistry?).
 - Ring conformations and planarity of aromatics and amides.
2. Run automated checks:
 - Validate valence, bond orders, and charges.
 - Detect severe steric clashes or atypical geometries.

12 Conclusions and Best-Practice Recommendations

Robust small-molecule preparation is a prerequisite for meaningful docking-based virtual screening. Key recommendations are:

- **Treat ligand preparation as a modeling step, not a trivial pre-processing task.** Empirical studies show that preparation parameters can substantially affect enrichment.[11, 1]
- **Use chemically informed protonation and tautomer states.** Incorporate pK_a prediction and tautomer enumeration; avoid assuming a single arbitrary state for multiprotic systems.
- **Enumerate relevant stereochemistry.** Do not ignore undefined stereocenters if stereochemistry is expected to affect binding.
- **Employ validated conformer generators.** Use tools such as OMEGA and RDKit/ETKDG, which have been benchmarked against crystallographic data.[? ?]
- **Filter out chemically implausible and highly strained conformers.** Apply energetic and geometric criteria to remove outliers.
- **Leverage curated databases.** Build libraries from ZINC, ChEMBL, PubChem, PDBbind, and established benchmarks (DUD-E, DEKOIS 2.0) to maximize data quality and facilitate comparison with literature.[8, 3? ? ?]
- **Verify.** Combine automatic checks with manual inspection of representative ligands and docked poses to ensure that the modeling remains chemically reasonable and that docking is “right for the right reasons.”[?]

As docking and machine-learning-based scoring functions continue to evolve, the importance of high-quality ligand preparation only increases. In a data-rich era, careful attention to protonation, tautomerism, stereochemistry, and conformational variability remains essential to converting raw compound collections into physically meaningful, docking-ready libraries.

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